```python
# Program to generate and plot look-up tables
# Made by: Neo Vorsatz

import numpy as np
from matplotlib import pyplot as plt
from scipy.io import wavfile

F_SIGNAL = 0.5
PLOT = "drum"
NS = 1024
MAX = 4095

#generate sine LUT
lut = []
file = open(file="sine_lut.txt", mode="w")
file.write("{")
for t in range(NS):
    val = round((MAX/2)*(1+np.sin(2*np.pi*(t/NS))))
    lut.append(val)
    file.write(str(val))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="sine":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Sine Wave LUT")
    plt.show()

#generate sawtooth LUT
lut = []
file = open(file="sawtooth_lut.txt", mode="w")
file.write("{")
for t in range(NS):
    val = round(t*(MAX/NS))
    lut.append(val)
    file.write(str(val))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="sawtooth":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Sawtooth Wave LUT")
    plt.show()
```

```python
#generate triangular LUT
lut = []
file = open(file="triangular_lut.txt", mode="w")
file.write("{")
for t in range(NS):
    if t<NS/2:
        val = round(2*t*(MAX/NS))
    else:
        val = round(MAX-2*(t-NS/2)*(MAX/NS))
    lut.append(val)
    file.write(str(val))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="triangular":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Triangular Wave LUT")
    plt.show()

#generate LUT from WAV
def wav_to_lut(path:str) -> list:
    """
    Takes the file path to a .wav file, and returns the LUT
    """
    #calculate the number of samples
    frequency = 44100 #Hertz
    desired_period = 2 #seconds
    cropped_samples = frequency*desired_period #number of samples
    #extract the desired number of data points
    sample_rate, data = wavfile.read(path)
    if data.ndim > 1:
        data = data[:, 0] #get only the first channel
    data = data[:cropped_samples] #extracting the first few samples
    data = data.astype(np.float32) #convert type
    indices = np.linspace(0, len(data)-1, NS).astype(int)
    lut = data[indices] #extract 128 values

    #normalise and round
    lut -= np.min(lut) #shift lowest value to 0
    lut *= MAX/np.max(lut) #normalise to the maximum
    lut = lut.astype(int)

    return lut

#generate piano LUT
lut = wav_to_lut("WAV_FILES/piano.wav")
file = open(file="piano_lut.txt", mode="w")
file.write("{")
```

```python
for t in range(NS):
    file.write(str(lut[t]))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="piano":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Piano LUT")
    plt.show()

#generate guitar LUT
lut = wav_to_lut("WAV_FILES/guitar.wav")
file = open(file="guitar_lut.txt", mode="w")
file.write("{")
for t in range(NS):
    file.write(str(lut[t]))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="guitar":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Guitar LUT")
    plt.show()

#generate drum LUT
lut = wav_to_lut("WAV_FILES/drum.wav")
file = open(file="drum_lut.txt", mode="w")
file.write("{")
for t in range(NS):
    file.write(str(lut[t]))
    if t==NS-1:
        file.write("}")
    else:
        file.write(",")
file.close()

if PLOT=="drum":
    plt.plot(range(NS), lut, color="black")
    plt.xlabel("Time")
    plt.ylabel("Amplitude")
    plt.title("Drum LUT")
    plt.show()
```

/* USER CODE BEGIN Header /
/*

- @file : main.c
- @brief : Main program body

- @attention
-
-
-
-
-
-
-
-

/
/ USER CODE END Header /
/ Includes --*/
#include "main.h"

/* Private includes --/
/ USER CODE BEGIN Includes /
#include <stdio.h>
#include "stm32f4xx.h"
#include "lcd_stm32f4.h"
/ USER CODE END Includes */

/* Private typedef ---/
/ USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define --/
/ USER CODE BEGIN PD */
// TODO: Add values for below variables
#define NS 1024 // Number of samples in LUT
#define TIM2CLK 16000000 // STM Clock frequency: Hint You might want to check the ioc file
#define F_SIGNAL 0.5 // Frequency of output analog signal

/* USER CODE END PD */

/* Private macro ---/
/ USER CODE BEGIN PM */

```
/* USER CODE END PM */

/* Private variables ---------------------------------------------------------*/
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim3;
DMA_HandleTypeDef hdma_tim2_ch1;

/* USER CODE BEGIN PV */
// TODO: Add code for global variables, including LUTs
const uint32_t Sin_LUT[NS] =
{2048,2060,2073,2085,2098,2110,2123,2135,2148,2161,2173,2186,2198,2211,2223,2236,2248,2261,2273,2286,229
8,2311,2323,2335,2348,2360,2373,2385,2398,2410,2422,2435,2447,2459,2472,2484,2496,2508,2521,2533,2545,25
57,2569,2581,2594,2606,2618,2630,2642,2654,2666,2678,2690,2702,2714,2725,2737,2749,2761,2773,2784,2796,2
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3046,3057,3068,3078,3089,3100,3111,3122,3132,3143,3154,3164,3175,3185,3195,3206,3216,3226,3237,3247,3257
,3267,3277,3287,3297,3307,3317,3327,3337,3346,3356,3366,3375,3385,3394,3404,3413,3423,3432,3441,3450,345
9,3468,3477,3486,3495,3504,3513,3522,3530,3539,3548,3556,3565,3573,3581,3590,3598,3606,3614,3622,3630,36
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791,3797,3804,3810,3816,3823,3829,3835,3841,3847,3853,3859,3865,3871,3876,3882,3888,3893,3898,3904,3909,
3914,3919,3924,3929,3934,3939,3944,3949,3953,3958,3962,3967,3971,3975,3980,3984,3988,3992,3996,3999,400
3,4007,4010,4014,4017,4021,4024,4027,4031,4034,4037,4040,4042,4045,4048,4051,4053,4056,4058,4060,4063,40
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4087,4086,4085,4084,4083,4081,4080,4078,4076,4075,4073,4071,4069,4067,4065,4063,4060,4058,4056,4053,405
1,4048,4045,4042,4040,4037,4034,4031,4027,4024,4021,4017,4014,4010,4007,4003,3999,3996,3992,3988,3984,39
80,3975,3971,3967,3962,3958,3953,3949,3944,3939,3934,3929,3924,3919,3914,3909,3904,3898,3893,3888,3882,3
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1,691,682,672,663,654,645,636,627,618,609,600,591,582,573,565,556,547,539,530,522,514,505,497,489,481,473,4
65,457,449,441,433,426,418,410,403,395,388,381,373,366,359,352,345,338,331,324,318,311,304,298,291,285,279,
272,266,260,254,248,242,236,230,224,219,213,207,202,197,191,186,181,176,171,166,161,156,151,146,142,137,13
3,128,124,120,115,111,107,103,99,96,92,88,85,81,78,74,71,68,64,61,58,55,53,50,47,44,42,39,37,35,32,30,28,26,24,
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26,28,30,32,35,37,39,42,44,47,50,53,55,58,61,64,68,71,74,78,81,85,88,92,96,99,103,107,111,115,120,124,128,133,1
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2,1985,1997,2010,2022,2035};
const uint32_t Saw_LUT[NS] =
{0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60,64,68,72,76,80,84,88,92,96,100,104,108,112,116,120,124,128,132,13
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44,1248,1252,1256,1260,1264,1268,1272,1276,1280,1284,1288,1292,1296,1300,1304,1308,1312,1316,1320,1324,1
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83,3987,3991,3995,3999,4003,4007,4011,4015,4019,4023,4027,4031,4035,4039,4043,4047,4051,4055,4059,4063,4
067,4071,4075,4079,4083,4087,4091};
const uint32_t Triangle_LUT[NS] =
{0,8,16,24,32,40,48,56,64,72,80,88,96,104,112,120,128,136,144,152,160,168,176,184,192,200,208,216,224,232,240
,248,256,264,272,280,288,296,304,312,320,328,336,344,352,360,368,376,384,392,400,408,416,424,432,440,448,45
6,464,472,480,488,496,504,512,520,528,536,544,552,560,568,576,584,592,600,608,616,624,632,640,648,656,664,6
72,680,688,696,704,712,720,728,736,744,752,760,768,776,784,792,800,808,816,824,832,840,848,856,864,872,880,
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1669,2657,1685,1448,2166,1517,2693,1622,1618,1995,1260,2789,1320,1931,1809,1281,2636,1452,2104,1707,134
7,2441,1566,2342,1659,1399,2420,1582,2507,1715,1477,2264,1508,2653,1572,1554,2179,1349,2672,1450,1740,19
36,1311,2598,1413,1974,1756,1349,2533,1494,2209,1755,1450,2355,1672,2333,1729,1466,2334,1573,2526,1669,1
488,2262,1413,2674,1420,1728,1970,1369,2601,1404,1931,1789,1463,2461,1543,2039,1799,1433,2419,1608,2208,
1826,1372,2474,1540,2483,1643,1507,2267,1454,2603,1525,1657,1970,1463,2594,1449,1773,1980,1527,2262,220
7,1649,1533,1517,2729,1376,2539,1564,1255,2376,1869,2341,1847,1378,2126,1587,2683,1686,1565,2013,1310,26
87,1515,1994,1632,1588,2272,1823,1709,2078,1175,2469,1621,2210,1632,1606,2119,1759,2206,1891,1284,2543,1
300,2695,1421,1770,1787,1772,2229,1803,1535,2057,1381,2561,1599,1925,1881,1503,2404,1655,2097,1827,1542,
2266,1780,2132,1983,1136,2322,1413,2726,1938,1007,2296,2049,1817,2430,2660,1003,1626,2383,1345,1113,1224
,2722,1895,43,163,2135,1379,1897,720,1878,1038,1379,1212,834,3011,2250,2142,2307,2041,3555,2238,1966,203
0,1832,3300,1576,1839,1536,1532,2969,1426,1663,1851,1374,3150,1204,2241,1610,1035,2082,1242,1901,2508,67
8,2097,713,1845,1880,1255,1902,2163,2342,2907,1745,2179,2095,2458,2033,1943,1423,2156,2199,1943,2173};
const uint32_t Drum_LUT[NS] =
{2047,2047,2047,2047,2048,2046,2045,2042,2050,2050,2048,2049,2042,1855,2239,2632,2118,669,4090,68,1563,4
091,0,2681,4088,798,1275,2260,371,1433,1341,1523,2104,3611,4058,3881,2630,2188,1702,2,0,927,2330,2509,256
2,3313,3252,3330,3295,1541,484,1080,1712,2285,2368,1778,2069,2174,2108,2405,2263,2158,2246,2033,1457,107
3,1579,1962,2241,2694,2703,2369,2517,2493,1943,1592,1375,1355,1692,1928,2195,2621,2638,2306,2072,1890,18
45,1930,2061,2153,1979,1867,1889,1872,1957,2030,2169,2449,2513,2409,2147,1719,1495,1476,1741,2015,2296,2
411,2434,2368,2304,1928,1715,1679,1773,2038,2218,2249,2193,2062,1947,1887,1881,2054,2242,2291,2214,2095,
1934,1822,1801,1876,1987,2141,2240,2273,2226,2105,2026,1941,1870,1868,1972,2082,2157,2168,2116,2039,195
7,1950,2011,2076,2124,2133,2099,2049,1993,1953,1957,1998,2065,2118,2111,2079,2072,2060,2032,2010,1982,19
96,2021,2042,2064,2099,2122,2116,2084,2027,1984,1976,1987,2018,2048,2075,2087,2079,2059,2051,2036,2032,2
046,2062,2103,2051,1807,1977,1722,2488,2530,2643,0,4084,1922,8,4091,3087,603,2797,872,479,1843,1756,1827,
2930,3892,4095,3287,2176,2328,80,0,6,1711,2655,2406,2814,3400,3212,3500,2669,762,655,1442,1937,2345,2046,
1876,2215,2084,2256,2417,2193,2176,2194,1811,1172,1261,1808,2031,2473,2808,2531,2394,2570,2242,1741,1483
,1303,1488,1829,2003,2410,2715,2494,2185,1978,1833,1894,1979,2147,2091,1890,1869,1881,1893,1997,2089,231
2,2468,2394,2353,1900,1661,1444,1620,1874,2135,2361,2467,2387,2364,2129,1814,1688,1683,1863,2155,2260,22
27,2136,1984,1908,1880,1939,2162,2283,2264,2161,2005,1871,1808,1815,1927,2060,2201,2267,2260,2181,2062,1
978,1904,1854,1906,2020,2127,2175,2149,2086,2001,1941,1970,2038,2105,2134,2119,2085,2023,1973,1958,1985,
2032,2096,2123,2096,2074,2069,2049,2021,1995,1987,2009,2030,2048,2081,2108,2123,2103,2060,2006,1976,198
0,2001,2031,2057,2082,2088,2074,2053,2041,2027,2036,2061,2085,2063,1995,816,4095,4,3874,3787,8,1005,4077,
11,1367,3566,1468,1867,2277,2853,2400,2518,2036,2130,1397,1576,2414,2183,1809,958,1214,2614,2431,1240,25
58,2495,2208,1721,2755,1794,2090,2208,1757,1884,2422,1898,1613,2144,2261,1738,2018,2132,1648,2013,2721,1
837,1986,2410,1888,2101,2324,1887,1999,2256,1909,1981,2245,2093,1916,2122,2046,1910,2075,2113,1931,2054,
2093,1960,2023,2109,2053,2010,2066,2072,2014,2078,2078,2007,2099,2131,2044,2020,2059,2014,1983,2039,201
5,2020,2095,2083,2059,2075,2111,2051,2038,2050,2006,2033,2050,2063,2018,2035,2080,2028,2054,2057,2072,20
44,2091,2036,2067,2068,2022,2019,2046,2040,2039,2051,2062,2032,2048,2016,2032,2043,2066,2046,2044,2054,2
040,2042,2048,2024,2045,2051,2051,2043,2045,2041,2049,2056,2053,2043,2064,2049,2040,2048,2053,2045,2046,
2047,2037,2044,2041,2047,2046,2052,2040,2038,2051,2049,2041,2035,2045,2051,2044,2027,2050,2045,2036,204
4,2055,2058,2049,2039,2079,1470,3823,3,1105,4095,0,2316,4086,1094,1450,1558,407,2020,1499,1433,2500,3569,
3878,4004,2489,2171,1287,1,0,1194,2801,2616,2572,3439,2968,3505,3366,1082,46,973,1501,2167,2520,1935,219
3,2342,2458,2293,2321,2064,2103,1924,1321,911,1698,2029,2730,3267,2575,2168,2419,2179,1722,1399,1374,147
6,1856,1998,2165,2585,2714,2434,2231,1783,1733,1843,2095,2076,1840,1691,1693,2041,2459,2510,2431,2477,21
```

42,2034,1660,1361,1524,1804,2297,2453,2464,2287,2025,1889,1830,1986,2000,2085,2142,2180,2077,2066,1859,1
714,1736,1969,2245,2455,2553,2248,1978,1796,1644,1803,1900,2108,2225,2414,2260,2007,1906,1814,1862,1946,
2144,2126,2103,2211,2132,2080,1832,1859,1982,2110,2070,2095,2144,2176,2046,1873,1915,1979,2035,2132,2063
,2082,2077,1991,1927,1969,2082,2025,1999,2038,2109,2135,2040,2071,2080,1996,1929,1984,2028,2071,2044,210
0,2082,2152,2043,2022,2015,1989,1985,2059,2013,2075,2057,2067,2022,2039,2090,2168,2076,2343,1732,2105,22
58,1785,2278,1974,2132,1869,2198,2260,1694,2030,2007,2075,1798,1912,2186,1991,1897,2093,2179,2192,2199,2
259,1809,2023,2091,2032,2090,2054,1969,2059,2091,2049,2151,1906,2085,1958,2041,2080,2018,2117,2150,2018,
2043,1998,2031,2096,1984,2044,2075,2022,2092,2017,2039,2090,2015,2050,2040,2046,2059,2051,2045,2075,203
0,2039,2036,2017,2058,2048,2062,2082,2054,2057,2040,2033,2040,2046,2053,2049,2036,2053,2035,2034,2046,19
88,2073,1926,2045,2063,2030,2090,2043,2050,2045,2054,2035,2101,1989,2079,2071,2019,2050,2046,2063,2026,2
040,2033,2065,2039,2054,2071,2047,2063,2040,2037,2053,2043,2041,2056,2027,2058,2032,2033,2047,2047,2059,
2067,2063,2069,2052,2044,2040,2031,2034,2030,2039,2041,2052,2049,2054,2053,2054,2053,2051,2045,2051,204
7,2042,2043,2043,2041,2041,2048,2047,2052,2049,2046,2051,2045,2048,2047,2049,2048,2048,2042,2042,2041,20
42,2044,2047,2049,2056,2051,2054,2044,2043,2046,2041,2057,2329,2065,1811,2299,2161,1975,2572,1944,2470,2
129,1472,1954,2326,2014,1834,2163,1761,2364,1901,2346,2183,2079,1691,2257,1773,1728,2093,1970,1995,1983,
1976,2091,1921,2026,2245,1971,1926,2158,2052,2128,1909,2038,2008,2006,2077,1990,2021,2105,2013,2029,211
2,2066,2101,2075,1974,2110,2040,1972,2129,2087,2096,2133,2058,2041,2040,2046,2084,2059,2037,2062,2028,20
56,2029,2053,2046,2050,2044,2058,2047,2054,2042,2039,2042,2047,2042,2047,2052,2047,2039,2113,2040,2069,2
048,2076,2053,2030,2036,2061,2039,2043,1999,2068,2052,2044,2035,2033};
uint32_t prev_tick = 0;
uint32_t waveform = 0;

// TODO: Equation to calculate TIM2_Ticks
uint32_t TIM2_Ticks = TIM2CLK/(F_SIGNAL*NS); // How often to write new LUT value
uint32_t DestAddress = (uint32_t) &(TIM3->CCR3); // Write LUT TO TIM3->CCR3 to modify PWM duty cycle

/* USER CODE END PV */

/* Private function prototypes ---/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_DMA_Init(void);
static void MX_TIM2_Init(void);
static void MX_TIM3_Init(void);
/ USER CODE BEGIN PFP /
void EXTI0_IRQHandler(void);
/ USER CODE END PFP */

/* Private user code ---/
/ USER CODE BEGIN 0 */

/* USER CODE END 0 */

/**

- @brief The application entry point.
- @retval int
  */
  int main(void)
  {

```
/* USER CODE BEGIN 1 */

/* USER CODE END 1 */

/* MCU Configuration--------------------------------------------------------*/

/* Reset of all peripherals, Initializes the Flash interface and the Systick. */
HAL_Init();

/* USER CODE BEGIN Init */

/* USER CODE END Init */

/* Configure the system clock */
SystemClock_Config();

/* USER CODE BEGIN SysInit */

/* USER CODE END SysInit */

/* Initialize all configured peripherals /
MX_GPIO_Init();
MX_DMA_Init();
MX_TIM2_Init();
MX_TIM3_Init();
/ USER CODE BEGIN 2 /
// TODO: Start TIM3 in PWM mode on channel 3;
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_3);
TIM3->CCR3 = (1<<15);
// TODO: Start TIM2 in Output Compare (OC) mode on channel 1
HAL_TIM_OC_Start(&htim2, TIM_CHANNEL_1);
// TODO: Start DMA in IT mode on TIM2->CH1. Source is LUT and Dest is TIM3->CCR3; start with Sine LUT
HAL_DMA_Start_IT(&hdma_tim2_ch1, Sin_LUT, DestAddress, NS);
// TODO: Write current waveform to LCD(Sine is the first waveform)
init_LCD();
lcd_command(CLEAR);
lcd_update_line("Sine");
// TODO: Enable DMA (start transfer from LUT to CCR)
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
/ USER CODE END 2 */

/* Infinite loop /
/ USER CODE BEGIN WHILE /
while (1)
{
/ USER CODE END WHILE */


  /* USER CODE BEGIN 3 */


}
/* USER CODE END 3 */
```

}

/**

- @brief System Clock Configuration
- @retval None
  */
  void SystemClock_Config(void)
  {
  RCC_OscInitTypeDef RCC_OscInitStruct = {0};
  RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

/** Configure the main internal regulator output voltage
*/
__HAL_RCC_PWR_CLK_ENABLE();
__HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE3);

/** Initializes the RCC Oscillators according to the specified parameters

- in the RCC_OscInitTypeDef structure.
  */
  RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
  RCC_OscInitStruct.HSIState = RCC_HSI_ON;
  RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
  RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;
  if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
  {
  Error_Handler();
  }

/** Initializes the CPU, AHB and APB buses clocks
*/
RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
|RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_0) != HAL_OK)
{
Error_Handler();
}
}

/**

- @brief TIM2 Initialization Function
- @param None

- @retval None
  */
  static void MX_TIM2_Init(void)
  {

/* USER CODE BEGIN TIM2_Init 0 */

/* USER CODE END TIM2_Init 0 */

TIM_ClockConfigTypeDef sClockSourceConfig = {0};
TIM_MasterConfigTypeDef sMasterConfig = {0};
TIM_OC_InitTypeDef sConfigOC = {0};

/* USER CODE BEGIN TIM2_Init 1 */

/* USER CODE END TIM2_Init 1 /
htim2.Instance = TIM2;
htim2.Init.Prescaler = 0;
htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
// htim2.Init.Period = 4294967295;
htim2.Init.Period = TIM2_Ticks;//#
htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
{
Error_Handler();
}
sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
{
Error_Handler();
}
if (HAL_TIM_OC_Init(&htim2) != HAL_OK)
{
Error_Handler();
}
sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
{
Error_Handler();
}
sConfigOC.OCMode = TIM_OCMODE_TIMING;
sConfigOC.Pulse = 0;
sConfigOC.OCPolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
if (HAL_TIM_OC_ConfigChannel(&htim2, &sConfigOC, TIM_CHANNEL_1) != HAL_OK)
{
Error_Handler();

}
/ USER CODE BEGIN TIM2_Init 2 /
/ TIM2_CH1 DMA Init */
__HAL_RCC_DMA1_CLK_ENABLE();

hdma_tim2_ch1.Instance = DMA1_Stream5;
hdma_tim2_ch1.Init.Channel = DMA_CHANNEL_3; // TIM2_CH1 is on channel 3
hdma_tim2_ch1.Init.Direction = DMA_MEMORY_TO_PERIPH; // Memory -> TIM3->CCR3
hdma_tim2_ch1.Init.PeriphInc = DMA_PINC_DISABLE; // Peripheral address fixed
hdma_tim2_ch1.Init.MemInc = DMA_MINC_ENABLE; // Memory address increments
hdma_tim2_ch1.Init.PeriphDataAlignment = DMA_PDATAALIGN_WORD;
hdma_tim2_ch1.Init.MemDataAlignment = DMA_MDATAALIGN_WORD;
hdma_tim2_ch1.Init.Mode = DMA_CIRCULAR; // Repeat LUT automatically
hdma_tim2_ch1.Init.Priority = DMA_PRIORITY_HIGH;
hdma_tim2_ch1.Init.FIFOMode = DMA_FIFOMODE_DISABLE;

if (HAL_DMA_Init(&hdma_tim2_ch1) != HAL_OK)
{
Error_Handler();
}

/* Link DMA handle to TIM2 handle /
__HAL_LINKDMA(&htim2, hdma[TIM_DMA_ID_CC1], hdma_tim2_ch1);
/ USER CODE END TIM2_Init 2 */

}

/**

- @brief TIM3 Initialization Function
- @param None
- @retval None
  */
  static void MX_TIM3_Init(void)
  {

/* USER CODE BEGIN TIM3_Init 0 */

/* USER CODE END TIM3_Init 0 */

TIM_ClockConfigTypeDef sClockSourceConfig = {0};
TIM_MasterConfigTypeDef sMasterConfig = {0};
TIM_OC_InitTypeDef sConfigOC = {0};

/* USER CODE BEGIN TIM3_Init 1 */

/* USER CODE END TIM3_Init 1 /
htim3.Instance = TIM3;
htim3.Init.Prescaler = 0;
htim3.Init.CounterMode = TIM_COUNTERMODE_UP;
// htim3.Init.Period = 65535;

```
htim3.Init.Period = (1<<12)-1;//#
htim3.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim3.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
if (HAL_TIM_Base_Init(&htim3) != HAL_OK)
{
Error_Handler();
}
sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
if (HAL_TIM_ConfigClockSource(&htim3, &sClockSourceConfig) != HAL_OK)
{
Error_Handler();
}
if (HAL_TIM_PWM_Init(&htim3) != HAL_OK)
{
Error_Handler();
}
sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
if (HAL_TIMEx_MasterConfigSynchronization(&htim3, &sMasterConfig) != HAL_OK)
{
Error_Handler();
}
sConfigOC.OCMode = TIM_OCMODE_PWM1;
sConfigOC.Pulse = 0;
// sConfigOC.Pulse = 1;//#
sConfigOC.OCPolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
if (HAL_TIM_PWM_ConfigChannel(&htim3, &sConfigOC, TIM_CHANNEL_3) != HAL_OK)
{
Error_Handler();
}
/ USER CODE BEGIN TIM3_Init 2 */

/* USER CODE END TIM3_Init 2 */
HAL_TIM_MspPostInit(&htim3);

}

/**
```

- Enable DMA controller clock
  */
  static void MX_DMA_Init(void)
  {

```
/* DMA controller clock enable */
__HAL_RCC_DMA1_CLK_ENABLE();
```

/* DMA interrupt init /
/ DMA1_Stream5_IRQn interrupt configuration */
HAL_NVIC_SetPriority(DMA1_Stream5_IRQn, 0, 0);
HAL_NVIC_EnableIRQ(DMA1_Stream5_IRQn);

}

/**

- @brief GPIO Initialization Function
- @param None
- @retval None
  /
  static void MX_GPIO_Init(void)
  {
  GPIO_InitTypeDef GPIO_InitStruct = {0};
  / USER CODE BEGIN MX_GPIO_Init_1 */

/* USER CODE END MX_GPIO_Init_1 */

/* GPIO Ports Clock Enable */
__HAL_RCC_GPIOH_CLK_ENABLE();
__HAL_RCC_GPIOA_CLK_ENABLE();
__HAL_RCC_GPIOB_CLK_ENABLE();

// ------------------------------
// LCD pins configuration
// ------------------------------
// Configure PC14 (RS) and PC15 (E) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_14 | GPIO_PIN_15;
GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
GPIO_InitStruct.Pull = GPIO_NOPULL;
GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);

// Configure PB8 (D4) and PB9 (D5) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_8 | GPIO_PIN_9;
HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

// Configure PA12 (D6) and PA15 (D7) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_12 | GPIO_PIN_15;
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Set all LCD pins LOW initially
HAL_GPIO_WritePin(GPIOC, GPIO_PIN_14 | GPIO_PIN_15, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8 | GPIO_PIN_9, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12 | GPIO_PIN_15, GPIO_PIN_RESET);

// ------------------------------
// Button0 configuration (PA0)
// ------------------------------

```
GPIO_InitStruct.Pin = Button0_Pin;
GPIO_InitStruct.Mode = GPIO_MODE_IT_RISING; // Interrupt on rising edge
GPIO_InitStruct.Pull = GPIO_PULLUP; // Use pull-up resistor
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Enable and set EXTI line 0 interrupt priority
HAL_NVIC_SetPriority(EXTI0_IRQn, 2, 0);
HAL_NVIC_EnableIRQ(EXTI0_IRQn);

/* USER CODE END MX_GPIO_Init_2 */

}

/* USER CODE BEGIN 4 */
void EXTI0_IRQHandler(void){

// TODO: Debounce using HAL_GetTick()
if (HAL_GetTick()-prev_tick>500) { //if more than 0.5 seconds have passed
prev_tick = HAL_GetTick();

// TODO: Disable DMA transfer and abort IT, then start DMA in IT mode with new LUT and re-enable transfer
// HINT: Consider using C's "switch" function to handle LUT changes
__HAL_TIM_DISABLE_DMA(&htim2,TIM_DMA_CC1);
HAL_DMA_Abort_IT(&hdma_tim2_ch1);
waveform = (waveform+1)%6;
switch (waveform) {
case 0:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Sin_LUT, DestAddress, NS);
lcd_update_line("Sine");
break;
case 1:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Saw_LUT, DestAddress, NS);
lcd_update_line("Sawtooth");
break;
case 2:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Triangle_LUT, DestAddress, NS);
lcd_update_line("Triangular");
break;
case 3:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Piano_LUT, DestAddress, NS);
lcd_update_line("Piano");
break;
case 4:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Guitar_LUT, DestAddress, NS);
lcd_update_line("Guitar");
break;
case 5:
HAL_DMA_Start_IT(&hdma_tim2_ch1, Drum_LUT, DestAddress, NS);
lcd_update_line("Drum");
```

break;
}
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
}

HAL_GPIO_EXTI_IRQHandler(Button0_Pin); // Clear interrupt flags
}
/* USER CODE END 4 */

/**

- @brief This function is executed in case of error occurrence.
- @retval None
  /
  void Error_Handler(void)
  {
  / USER CODE BEGIN Error_Handler_Debug /
  / User can add his own implementation to report the HAL error return state /
  __disable_irq();
  while (1)
  {
  }
  / USER CODE END Error_Handler_Debug /
  }
  #ifdef USE_FULL_ASSERT
  /
- @brief Reports the name of the source file and the source line number
- ```
      where the assert_param error has occurred.
  ```
- @param file: pointer to the source file name
- @param line: assert_param error line source number
- @retval None
  */
  void assert_failed(uint8_t *file, uint32_t line)*
  {
  / USER CODE BEGIN 6 /
  / User can add his own implementation to report the file name and line number,
  ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) /
  / USER CODE END 6 /
  }
  #endif / USE_FULL_ASSERT */
  ```